Also, let monthly expenses of Komal and Asha be 3y and 2y.

Now, $4x - 3y = 600$ .....(i)

$3x - 2y = 600$ .....(ii)

Solving (i) and (ii), x = 600 and y = 600

∴ Total monthly income = (4 + 3)(600) = Rs.4200

148. If X = –2, then

$X^3 - X^2 - X - 1$ is equal to

(a) 1 (b) –3

(c) –11 (d) –15

Ans. (c) : x = – 2

$= x^3 - X^2 - x - 1 = 0$

By putting x = – 2

$(-2)^3 - (2)^2 - (-2) - 1$

$= -8 - 4 + 2 - 1$

$= -11$

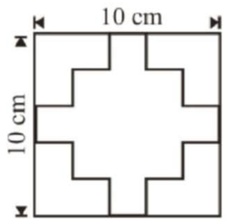
149. The Raga which is sung early in the morning is:

(a) Todi (b) Darbari

(c) Bhopali (d) Bhimpalasi

Ans. (a) : Raga Todi belongs to Todi Thaat. It is a morning Raga, and uses all seven nodes in the ascent and descent Rishabh. Gandhar and Dhaivat are komal (flat), Madhyam-is Tivra while Nishad is Shuddha. The derivative ragas out of this structure are grouped under the broud head of Todi Thaat.

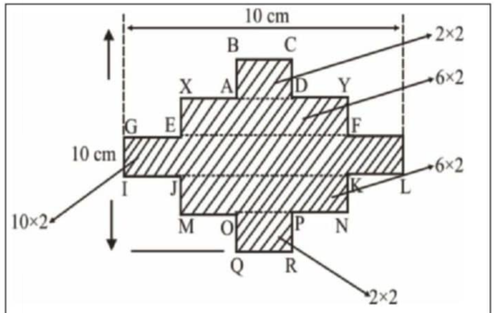
150. In the given figure, all line segments of the shaded portions are of the same length and at right angles to each other. The same can be cut out of a board of side 10 cm. What is the area of the shaded portion?



(a) 36 cm^2 (b) 48 cm^2

(c) 52 cm^2 (d) 64 cm^2

Ans. (c) :



According to the question all line segment are same length.

So GE + XA + BC + DY + FX = 10

GE = XA = BC = DY = FX = 2 cm each

Area of shaded position = (Area of ABCD + Area OPQR)

+ (Area EXDY + Area JMNK) + Area GILH

$[2 \times 2 + 2 \times 2] + [6 \times 2 + 6 \times 2] + [2 \times 10]$

$= 8 + 24 + 20 = 52\ cm^2$

Hence area of shaded portion = 2 (Area of square) + 2 (Area of rectangle EFGH IJKL) + Area of rectangle ABCD $= 2(4 + 6 \times 2) + 10 \times 2 = 52\ cm^2$.

UNION PUBLIC SERVICE COMMISSION

Civil Services (Preliminary Exam) - 1999

GENERAL STUDIES : PAPER-I

Time: 2 hours **Maximum Number: 200**

ANCIENT HISTORY

1. From the third century AD when the Hun invasion ended the Roman Empire, the Indian merchants relied more and more on the:

(a) African trade

(b) West-European trade (c) South-East Asian trade

(d) Middle-Eastern trade

Ans. (c) : After the 3rd century AD, Roman Empire came to an end. Indian merchants meanwhile had begun to rely more heavily on the south-east Asian Trade.

2. The following persons came to India at one time or another:

1. Fa-Hien 2. I-Tsing

3. Megasthenese 4. Hieum-Tsang

The correct chronological sequence of their visits is:

(a) 3, 1, 2, 4 (b) 3, 1, 4, 2

(c) 1, 3, 2, 4 (d) 1, 3, 4, 2

Ans. (b) : Fa-Hien – 405-411 CE

I-Tsing – 671-695 CE

Megasthanese – 302-298 BC

Hieun-Tsang – 630-645CE

3. Which one of the following was intitially the most powerful city state of India in the 6th century B.C.?

(a) Gandhar (b) Kamboj

(c) Kashi (d) Magadh

Ans. (d) : Of the 16 Mahajanapadas, Magadha, Kosala, Vatsa and Avanti were more powerful. They fought amongst themselves for years and ultimately Magadha emerged victorious under Bimbisara (Haranyak dynasty) during 6th Century BC.

4. The Indo-Greek Kingdom set up in north Afghanistan in the beginning of the second century B.C. was:

(a) Bactria (b) Scythia

(c) Zedrasia (d) Aria

Ans. (a) : Bactria was the ancient civilizations of Iranian peoples. Ancient Bactria was located between the Hindu Kush mountain range and the Amu Darya river, covering the flat region that straddles modern-day Afghanistan and Tajikistan.

5. One consistent feature found in the history of southern India was the growth of small regional kingdoms rather than large empires because of :

(a) The absence of minerals like iron

(b) Too many divisions in the social structure

(c) The absence of vast areas of fertile land

(d) The scarcity of manpower

Ans. (c) : One consistent feature found in the history of southern India was the growth of small regional kingdoms rather than large empires was due to the fact that in southern India, fertile lands were scattered and the population was centered around those fertile land. So due to the absence of large fertile land, small regional kingdoms were developed in southern India.

6. The term 'Aryan' denotes:

(a) An ethnic group

(b) A nomadic people

(c) A speech group

(d) A superior race

Ans. (c) : The term Aryan denotes a linguistic group and not a race. Their language known as Vedic Sanskrit has definite relationship with major languages of Europe and Asia. Scholars call this group of languages as Indo-European and the people speaking these languages as Indo-Europeans or Indo-Aryans.

7. Which one of the following ports handled the north Indian trade during the Gupta period?

(a) Tamralipti

(b) Broach

(c) Kalyan

(d) Cambray

Ans. (a) : Tamralipta or Tamralipti was the name of an ancient city on the Bay of Bengal corresponding with Tamluk in modern-day India. Tamralipta may had been one of the most important urban centres of trade and commerce of early historic India, trading along the Silk Road with China. Broach was the main port of western India where sea trade done with western countries.